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COURSE CODE/TITLE:	CPE010 - Data Structures and Algorithms	**SECTION:**	CPE21S1
DATE PERFORMED:	Oct 20, 2026	**DATE SUBMITTED:**	Oct 20, 2026
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ACTIVITY NO. _14_ [Algorithm Complexity]

1. PROCEDURE OUTPUT

What procedures did you perform?

	Best Case	Worst Case
Selection Sort	O(n^2)	O(n^2)
Bubble Sort	O(n^2)	O(n^2)
ANALYSIS		
Garcia	Selection Sort: The code uses nested loops that run it all through at any circumstance, even when it is already sorted and arranged it still checks every pair and it hits O(n^2) every single time. Worst Case: It calls RoutineSmallest to scan through every array in the loop, making it so it forces to do the same amount of comparisons whether the array is sorted or not.	

Table 14-1. Best- and Worst-Case Analysis using Theoretical Tools

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Best Case		
N	**Bubble Sort**	**Selection Sort**
N = 100	0.045 ms	0.022 ms
N = 1,000	3.280 ms	1.890 ms
N = 10,000	385.20 ms	182.40 ms

Worst case		
N	**Bubble Sort**	**Selection Sort**
N = 100	0.051 ms	0.024 ms
N = 1,000	4.150 ms	1.950 ms
N = 10,000	412.60 ms	189.10 ms

Table 14-2. Best- and Worst-Case Analysis using Experimental Tools

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Pseudocode

```
//Linear Search
Algorithm LinearSearch(arr, n, target)
    FOR i = 0 TO n - 1 DO
        IF arr[i] == target THEN
            RETURN i
        END IF
    END FOR
    RETURN -1
```

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```
//Binary Search
Algorithm BinarySearch(arr, n, target)
    low = 0
    high = n - 1
    WHILE low <= high DO
        mid = low + (high - low) / 2
        IF arr[mid] == target THEN
            RETURN mid
        ELSE IF arr[mid] < target THEN
            low = mid + 1
        ELSE
            high = mid - 1
        END IF
    END WHILE
    RETURN -1
```

Time and Space Complexities				
Searching Algorithm	**Best Case**	**Worst Case**	**Average Case**	**Auxiliary Space**
Linear Search	Omega 1	O(n)	O(n)	O(1)
Binary Search	Omega 1	O(logn)	O(logn)	O(1)

Input Size (N)	Elapsed Time for Binary Search (Worst-Case)	Elapsed Time for Linear Search (Worst-Case)
1,000	0.002 ms	0.028 ms
10,000	0.310 ms	0.310 ms
100,000	0.009 ms	3.420 ms

Conclusion

Through this activity, I learned how asymptotic notation works and how to determine an algorithm's growth rate using Big O, Big Omega, and Big Theta. By analyzing code loops and mathematical expressions, I realized that as the input size n grows, low order terms fade in significance compared to the highest power term, which dictates the overall growth class. Going through the procedure step by step and filling out the comparison tables made it much easier to visualize how fast an algorithm slows down when moving from linear to quadratic complexity, while the supplementary problems showed me how best and worst case scenarios can have completely different asymptotic bounds.

Overall, I think I did well in picking up the main concepts and identifying the upper, lower, and tight limits once I focused on the leading terms. I felt comfortable analyzing basic formulas and code snippets, but my main area for improvement is practicing more on formal mathematical proofs with constants and double-checking my steps whenever logarithmic terms like n \log n are involved.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.



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